BL-03 Database Schema

Backlog item: BL-03 (GitHub issue #3) | Sprint 1 | Database: PostgreSQL | Defined as Django models in `apps/*/models.py`, created by the migrations in `apps/*/migrations/`.

Diagram

```
erDiagram
    USER ||--o| STUDENT : "has profile"
    USER ||--o| FACULTY : "has profile"
    SECTION |o--o{ STUDENT : "home section of"
    SECTION ||--o{ TEACHING_ASSIGNMENT : offers
    SUBJECT ||--o{ TEACHING_ASSIGNMENT : "taught as"
    FACULTY |o--o{ TEACHING_ASSIGNMENT : teaches
    STUDENT ||--o{ ENROLLMENT : "enrolled in"
    TEACHING_ASSIGNMENT ||--o{ ENROLLMENT : has
    TEACHING_ASSIGNMENT ||--o{ ATTENDANCE_SESSION : "meets as"
    ATTENDANCE_SESSION ||--o{ ATTENDANCE_RECORD : contains
    STUDENT ||--o{ ATTENDANCE_RECORD : receives

    USER {
        bigint id PK
        string username UK "roll number or faculty ID"
        string password "salted hash only"
        string role "ADMIN | FACULTY | STUDENT"
        bool is_active "false = deactivated"
        bool must_change_password
        string email
    }
    STUDENT {
        bigint id PK
        bigint user_id FK,UK
        string roll_number UK
        string full_name
        bigint home_section_id FK "optional (BL-07)"
    }
    FACULTY {
        bigint id PK
        bigint user_id FK,UK
        string faculty_id UK
        string full_name
        string department
    }
    SUBJECT {
        bigint id PK
        string code UK
        string name
    }
    SECTION {
        bigint id PK
        string class_name "unique with name"
        string name
    }
    TEACHING_ASSIGNMENT {
        bigint id PK
        bigint section_id FK "unique with subject_id"
        bigint subject_id FK
        bigint faculty_id FK "optional"
    }
    ENROLLMENT {
        bigint id PK
        bigint student_id FK "unique with subject_id"
        bigint assignment_id FK
        bigint subject_id FK "copied from assignment"
    }
    ATTENDANCE_SESSION {
        bigint id PK
        bigint assignment_id FK "unique with date and period"
        date date
        int period "1 to 6"
        bigint recorded_by_id FK "optional"
        string recorded_by_code "faculty ID"
        datetime created_at
    }
    ATTENDANCE_RECORD {
        bigint id PK
        bigint session_id FK "unique with student_id"
        bigint student_id FK
        string status "P | A"
    }
    AUDIT_ENTRY {
        bigint id PK
        datetime timestamp
        bigint actor_user_id "not a foreign key"
        string actor_username
        string action
        string target_type
        string target_id
        json old_value
```

```
        json new_value
        text reason
        bool is_admin_override
    }
```

`AUDIT_ENTRY` has no links to other tables on purpose, so entries survive when users are deleted.

Tables

Table	App	Purpose
`accounts_user`	accounts	Login account and role. Username is the roll number or faculty ID (BL-01 D-02).
`academics_student`	academics	Student profile linked one-to-one to a user, with an optional home class/section (added in BL-07). Subject-by-subject membership is in `academics_enrollment`.
`academics_faculty`	academics	Faculty profile linked one-to-one to a user.
`academics_subject`	academics	Subject (course) with a unique code.
`academics_section`	academics	Class and section, e.g. "B.Tech CSE Year 2" + "A".
`academics_teachingassignment`	academics	A subject taught in a section, and the faculty member who teaches it.
`academics_enrollment`	academics	A student taking one subject in one section (enrolment is per subject, so electives are supported).
`attendance_attendancesession`	attendance	One class meeting.
`attendance_attendancerecord`	attendance	One student's status in one session.
`audit_auditentry`	audit	Append-only history of attendance and account changes.

Django also creates its standard tables for permissions, sessions, admin history and migrations.

Rules enforced by the database

Rule	How	Source
Role is Admin, Faculty or Student	Check constraint `user_role_valid`	FR-02
Usernames, roll numbers, faculty IDs and subject codes are unique	Unique constraints	UC-02, UC-03, UC-04
A section name is unique within its class	`section_unique_name`	UC-05
A subject is offered once per section	`assignment_unique_section_subject`	BL-11
A student is in only one section per subject	`enrollment_one_section_per_subject`	BL-01 D-19
Session key: section + subject + date + period	`session_unique_key` (the assignment identifies section + subject)	BL-01 D-08
Period is 1 to 6	`session_period_range`	BL-01 D-08, team decision (max 6 periods per day)
One record per student per session	`record_one_per_student_session`	BL-01 D-08
Status is Present (`P`) or Absent (`A`) only	`record_status_valid`	BL-01 D-06
Audit entries cannot be changed or deleted	PostgreSQL triggers reject UPDATE, DELETE and TRUNCATE	SEC-21, OBJ-04

What happens on deletion

Deleted	Effect	Source
Student (via their user account)	Profile, enrolments and that student's attendance records are deleted. Audit entries remain.	BL-01 D-04
Faculty member with teaching assignments	**Blocked** until the assignments are reassigned.	BL-01 D-03, D-04
Faculty member without assignments	Profile and login deleted; sessions they recorded remain and keep their faculty ID.	BL-01 D-04
Subject or section with attendance	**Blocked.**	BL-01 D-05
Subject or section without attendance	Its teaching assignments and enrolments are deleted.	FR-05
Section that is a student's home section	The student's home section is cleared; the student is kept.	BL-07
Audit entry	**Always blocked.**	SEC-21

Deactivation (BL-01 D-03) does not delete anything: it sets `is_active` to false on the user.

Rules checked by the application, not the database

These are enforced in later backlog items because they depend on the current date, settings or the user making the request:

- Session dates cannot be in the future; the correction window (BL-17).
- Every listed student must be marked before saving (BL-12, BL-13).

- Faculty may only mark their assigned classes (BL-05, BL-12).
- Threshold, notification state and correction-window settings are added in BL-22, BL-23 and BL-17.

Verification

BL-17 adds `attendance_correctionwindow` through migration `0004_correctionwindow`: one row (`id=1`), `days` constrained to 0-30, and `updated_at`. The migration inserts the default of two calendar days. This policy drives both faculty corrections and creation of missed sessions; Admin policy changes and attendance corrections use the existing append-only audit table. Attendance corrections update the existing record rather than creating a second student/session row.

`python manage.py test` runs 37 tests, including one for each rule above. The migrations were also applied to an empty database, and the audit trigger migration was reversed and re-applied successfully.